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**SUBJECT : OBJECT ORIENTED PROGRAMMING WITH
JAVA**

**TOPIC : LIMITATIONS OF TRADITIONAL ERROR
HANDLING**



TRADITIONAL ERROR HANDLING

- Early programming languages handled errors manually.
- Programmers used if-else conditions to check possible errors.
- Error messages or return values indicate failure.
- This method works for small programs but has several limitations.

EXAMPLE CODE:

```
➤ public class TraditionalErrorExample {  
➤     public static void main(String[] args) {  
  
➤         int a = 10;  
➤         int b = 0;  
  
➤         if(b != 0){  
➤             int result = a / b;  
➤             System.out.println("Result = " + result);  
➤         } else {  
➤             System.out.println("Error: Division by zero");  
➤         }  
  
➤     }  
➤ }
```



LIMITATION 1: CODE COMPLEXITY

- Large programs require many if-else conditions.
- Code becomes long and complex
- Difficult to read and maintain.
- Program logic becomes unclear.



LIMITATION 2: MIXING LOGIC WITH ERROR HANDLING

- Error handling code is mixed with normal program code.
- Makes the program difficult to understand.
- Hard to separate main logic and error checking.
- Example:

```
➤ if(file != null){  
➤     readFile();  
➤ }  
➤ else{  
➤     System.out.println("File not found");  
➤ }
```



LIMITATION 3 & 4

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- 1. Difficult Debugging

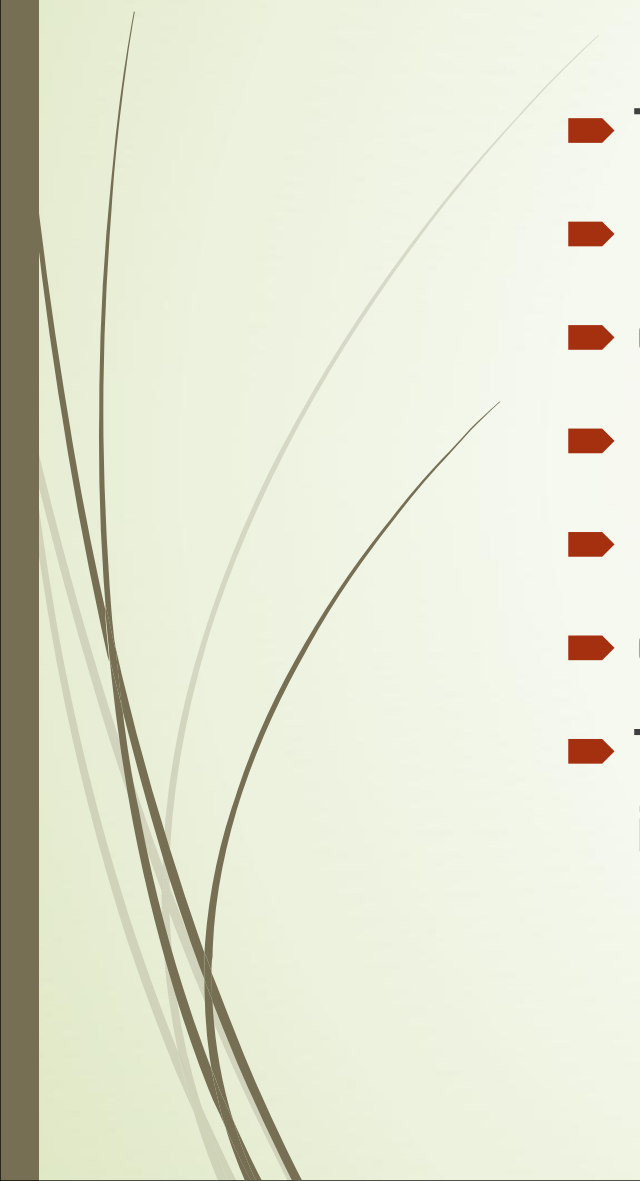
 - Errors are checked in many places.

 - Hard to locate where the actual error occurred.

- 2. Code Duplication

 - Same error checking code is repeated.

 - Increases program size unnecessarily

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- Traditional error handling uses conditional statements.
 - It has many limitations such as:
 - Complex code
 - Mixing logic and error handling
 - Difficult debugging
 - Code duplication
 - These problems led to the introduction of Exception Handling in Java.